O-C-105D
July 21, 1974
SUPERSEDING
Fed. Spec. O-C-105C
August 11, 1965

FEDERAL SPECIFICATION

CALCIUM CHLORIDE, DIHYDRATE AND CALCIUM CHLORIDE, ANHYDROUS: TECHNICAL

This specification was approved by the Commissioner, Federal Supply Services, General Services Administration, for use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for the technical grades of hydrated and anhydrous calcium chloride (see 6.1).

1.2 Classification. The calcium chloride covered by this specification shall be of the following types, grades, and classes, as specified, (see 6.2):

Type I - Dihydrate, 77 percent, flake.
Type II - Anhydrous.

Grade A - 94 percent, flake, pellet, or other granular.
Grade B - 90 percent.

Class 1 - Granular.
Class 2 - Lump.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

UU-S-48 - Sack, Shipping, Paper.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals.
PPP-D-723 - Drum, Fiber.
PPP-F-320 - Fiberboard: Corrugated and Solid, Sheet Stock (Container Grade, and Cut Shapes.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 - Quality Assurance Terms and Definitions.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM).

D345 - Methods of Sampling and Testing Calcium Chloride.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Chemical requirements. Calcium chloride shall not be caked or sticky and shall conform to the chemical requirements of Table I when tested as specified herein.

3.2 First article. When specified (see 6.2), the contractor shall furnish a one-pound sample of calcium chloride for first article inspection and approval (see 4.2.1).

TABLE I. CHEMICAL REQUIREMENTS

	Type I	Type II	
		Grade A	Grade B
Calcium chloride minimum, percent	77.0	94.0	90.0
Total alkali chloride (as NaCl), maximum, percent	2.0	5.0	4.2
Total magnesium (as $MgCl_2$), maximum, percent	0.5	0.5	0.5
Other impurities (not including water) maximum, percent	1.0	1.0	1.0

3.3 Particle size. The material shall conform to the requirements shown in Table II when tested as specified in 4.6.2.

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TABLE II. SIEVE ANALYSIS

Sieve size or number	Type I	Type II		
		Grade A	Grade B	
			Class 1	Class 2
Passing 1-inch (25.4 mm)	-	-	-	100 percent
Passing 3/8-inch (9.51 mm)	100 percent	100 percent	100 percent	-
Passing U.S. No. 4 (4.76mm)	90 percent, minimum	80 percent minimum	90 percent minimum	5 percent minimum
Passing U.S. No. 8 (2.38 mm)	-	-	50 percent, maximum	-
Passing U.S. No. 16 (1.19 mm)	-	-	5 percent maximum	-
Passing U.S. No. 20 (0.841 mm)	10 percent, maximum	10 percent, maximum	-	-

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection of materials and components. In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, examined, and tested to the extent specified, in accordance with the requirements of referenced subsidiary specifications and standards. The terms and definitions of MIL-STD-109 shall apply.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. First article inspection shall be performed on a one-pound sample of calcium chloride when a first article sample is required (see 3.2). This inspection shall include the examination of 4.5 and the tests of 4.6. Failure of the first article to pass the examination or any of the tests shall be casue for rejection. The first article may be extracted from the standard product in the supplier's current inventory provided the product meets the requirements of the specification and is representative of the chemical composition and manufacturing technique applicable to the remaining quantities to be furnished under the contract.

4.2.2 Quality conformance inspection. Quality conformance inspection shall be performed on the sample calcium chloride selected in accordance with 4.4. This inspection shall include the examination of 4.5 and the tests of 4.6.

4.3 Inspection lot. All units of the same type, grade, and class, offered to the Government at one time, shall be considered a lot, for purposes of inspection. The sample unit shall be one filled, primary container.

4.4 Sampling.

4.4.1 Sampling for examination. A random sample of filled calcium chloride containers shall be selected from each lot offered to the Government in accordance with MIL-STD-105 at inspection level S-3. The acceptable quality level shall be 1.5 percent for major defects and 2.5 percent for minor defects.

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4.4.2 Sampling for tests. Sampling for tests shall be performed in accordance with the method set forth in ASTM D345. Any sample tested in accordance with 4.6, that is rejected, shall be cause for rejection of the lot.

4.5 Examination. Each unit selected in accordance with 4.4.1 shall be examined to verify compliance with the requirements of this specification. Examination shall be conducted as specified in Table III.

4.6 Tests. Each sample selected in accordance with 4.4.2 shall be tested and any unit failing to pass the following tests, as applicable, shall be rejected. A certificate of compliance from an independent laboratory, approved by the contracting officer, will be accepted as evidence that the samples have passed the required tests. Tests shall be conducted as outlined in the referenced documents and as herein specified.

4.6.1 Test procedures. Tests shall be made in accordance with the applicable method of ASTM D345.

TABLE III. CLASSIFICATION OF DEFECTS

Categories	Defects
Major:	
101	Presence of dirt or foreign matter in calcium chloride.
102	Calcium chloride not homogeneous.
103	Calcium chloride not in form specified.
104	Objectionable odor.
Minor:	
201	Any evidence of leakage or seepage from any part of the container.
202	Any hole, tear, puncture, scuff, dent, or other defect in body of container.
203	Closures not as specified.
204	Printing or stenciling pertaining to the applicable information and instructions for end item use, of the specified type material omitted, illegible, incorrect, incomplete, or smeared.
205	Average net contents, less than the quantity specified.

4.7 Inspection of preparation for delivery. Preparation for delivery requirements of the calcium chloride shall be inspected to determine conformance with the applicable requirements of section 5 of this specifications.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 One-pound quantities. When specified (see 6.2), calcium chloride shall be packaged in one-pound quantities in glass bottles conforming to Group A, Class 1, Type d, Style 2, Grade 2, Closure A, Seal A of PPP-C-186.

5.1.1.2 Intermediate packaging. Calcium chloride units, packaged in accordance with 5.1.1.1 shall be intermediately packaged in accordance with the level A requirements of PPP-C-186.

5.1.1.3 Five-pound quantities. When specified (see 6.2), calcium chloride shall be packaged in five-pound quantities in glass bottles conforming to Group A, Class 1, Type d, Style 2, Grade 2, Closure A, Seal A of PPP-C-186. No intermediate packaging is required.

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5.1.2 Level C. Calcium chloride shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving agency.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 One-pound quantities. Calcium chloride, packaged in accordance with 5.1.1.1, shall be packed in accordance with the level A requirements of PPP-C-186.

5.2.1.2 Five-pound quantities. Calcium chloride, packaged in accordance with 5.1.1.3, shall be packed in three rows of four bottles per row in a close-fitting fiberboard box conforming to PPP-B-636, Type CF, Class weather-resistant, V3c. Separator pads and dividers of fiberboard conforming to PPP-F-320, W6, shall be used to separate the rows and the individual bottles.

5.2.1.3 Eighty- and one hundred-pound quantities. calcium chloride in eighty- or one hundred-pound quantities, as specified (see 6.2), shall be packed in packed in appropriate size sacks conforming to construction number 16X or 17X MB2 of UU-S-48, or fiber drums conforming to Type III, Grade A, of PPP-D-723, as specified (see 6.2). Fiber drums shall be provided with a polyethylene bag liner or other suitable water-vapor-proof liner or interior coating.

5.2.2 Level B.

5.2.2.1 One-pound quantites. Calcium chloride, packaged in accordance with 5.1.1.1 shall be packed in accordance with the level B requirements of PPP-C-186.

5.2.2.2 Five-pound quantites. Calcium chloride, packaged in accordance with 5.1.1.3 shall be packed in threeerows of four bottles per row in a close-fitting fiberboard box conforming to PPP-B-636, Type CF class domestic, grade 275. Separator pads and dividers of fiberboard conforming to PPP-F-320, W6, shall be used to separate the rows and the individual bottles.

5.2.2.3 Eighty- and one hundred-pound quantities. Calcium chloride in eighty- or one hundred-pound quantities, as specified (see 6.2), shall be packed in appropriate size sacks conforming to construction number 13-13X or 16-16X MB2 of UU-S-48, or fiber drums conforming to Type I, Grade A, of PPP-D-723, as specified (see 6.2). Fiber drums shall be provided with a polyethylene bag liner or other suitable water-vapor-proof linging or interior coating.

5.2.3 Level C. Calcium chloride shall be packed in quantities specified (see 6.2), in a manner which will insure arrival at destination in satisfactory condition and be acceptable to the carrier at lowest rates. Contianers and packing shall comply with the Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking. In addition to any special marking required in the procurement documents, marking of the interior apackages and shipping containers shall be in accordance with Fed. Std. No. 123 for civil agencies or MIL. STD. 129 for military agencies as applicable (see 6.2).

6. NOTES

6.1 Intended use. Type I and Type II, Grade A, calcium chloride, are intended for use as road construction material, admixtures for concrete, refrigerating mediums, and other technical uses. In addition, both types may be used for snow and ice control on sidewalks and driveways of buildings. Type II, Grade B, calcium chloride is intended for use as a desiccating (drying) agent. Calcium chloride should not be used as an antifreeze material in the watercooling system of internal combusion engines. The requirements for Type I and Type II, Grade A, calcium chloride, are essentially similar to the requirements of ASTM D98. CAUTION: Calcium chloride is a very hydroscopic material and should be stored in tightly closed containers.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.

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(b) Type, grade, and class of calcium chloride required (see 1.2).
(c) When first articlesis required for inspection and approval (see 3.2).
(d) Level of packaging and level of packing required (see 5.1 and 5.2).
(e) Quantity required per package (see 5.1.1.1 and 5.1.2).
(f) Quantity required per pack (see 5.2.1.2, 5.2.2.2,and 5.2.3).
(g) Type of pack required (see 5.2.1.2 and 5.2.2.2).
(h) Marking required (see 5.3).

Custodian:

Army - MU
Air Force - 68

PREPARING ACTIVITY:

Air Force - 68
Project No. 6810-B020

Review Activities:

Army - MU

CIVIL AGENCY COOPDINATING ACTIVITY:

GSA - FSS

User Activities:

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